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MULTIPLEXED SINGLE-PHOTON GENERATOR AND ASSOCIATED METHODS

Abstract

A multiplexed single-photon generator includes a plurality of single-photon sources, an optical switching network controllable to couple pump light into any one of the single-photon sources, and a bus waveguide optical coupled to all of the single-photon sources. Each single-photon source includes a quantum emitter coupled to an optical cavity. All of the single photons emitted by all of the single-photon sources propagate along the bus waveguide. The single-photon generator may be time-multiplexed in which only one of the single-photon sources is pumped at any time. In this case, the single photons form a temporal sequence with little or no temporal overlap. As an alternative to time multiplexing, the single-photon generator may be frequency-multiplexed such that each single-photon source emits single photons that are distinguishable from the other emitted single photons based on optical frequency.

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Background/Summary

RELATED APPLICATIONS [0001] This application claims priority to U.S. Provisional Patent Application No. 63/563,787, filed on Mar. 11, 2024, which is incorporated herein by reference in its entirety.

BACKGROUND

[0002] A single-photon source is a light source that emits single particles of light (i.e., single-photon Fock states) at specific wavelengths and specified times.

SUMMARY

[0003] Single-photon sources are important tools for many fields of science and engineering, including quantum optics, cryptography (e.g., quantum key distribution), quantum computing, and photonics. Also known as single-photon emitters, single-photon sources emit single photons that may be used as units of quantum information or as probes for various quantum sensors. For example, single-photon sources are frequently used in quantum optics to study the fundamental properties of light and how photons interact with matter. Single-photon sources also allow researchers to observe and manipulate individual photons, essential for studying phenomena like quantum entanglement, boson sampling, and quantum superposition. In another example, single-photon sources may be used to generate photonic qubits for quantum computing. In yet another example, single-photon sources may be used for high-speed, low-power optical communication systems. Single-photon sources may also be used for sensing applications, such as biomedical imaging and environmental monitoring.

[0004] The present embodiments include a multiplexed single-photon generator that combines multiple single-photon sources into a single platform. The single-photon generator includes an optical switching network that couples pump light into any one of the single-photon sources. The single-photon generator also includes a bus waveguide that is coupled to all of the single-photon sources so that all of the single photons emitted by all of the single-photon sources propagate in the same direction down the bus waveguide. Advantageously, the resulting stream of single photons has a photon rate (i.e., the number of single photons per unit time) that may be much higher than what any one single-photon source can generate alone. In addition, the single-photon generator may be fabricated using photonic integrated circuit (PIC) technologies, and therefore may be integrated with other photonic and electronic components onto a single chip.

[0005] In some embodiments, the single-photon generator is time-multiplexed in that only one of the single-photon sources is pumped at any time. The optical switching network may be controlled such that each single photon emitted does not exhibit any significant temporal overlap with other single photons in the stream. In these embodiments, the single photons may be indistinguishable with regards to frequency, i.e., all of the single-photon sources emit single photons at the same optical frequency. In other embodiments, the single-photon generator is frequency-multiplexed by configuring the single-photon sources to emit single photons at different frequencies. In this case, the single photons emitted by one of the single-photon sources are distinguishable from all other single photons based on optical frequency.

[0006] In some embodiments, a multiplexed single-photon generator includes a plurality of single-photon sources. Each of the plurality of single-photon sources includes an optical cavity and a quantum emitter coupled to the optical cavity. The multiplexed single-photon generator further

includes an optical switching network controllable to couple pump light into the optical cavity of any one of the plurality of single-photon sources. The multiplexed single-photon generator further includes a bus waveguide coupled to the optical cavity of each of the plurality of single-photon sources.

[0007] In some embodiments, a method for single-photon generation includes coupling a first pump pulse into the optical switching network of any multiplexed single-photon generator of the present embodiments. The method further includes controlling the optical switching network to couple the first pump pulse into the optical cavity of a first single-photon source of the plurality of single-photon sources of the multiplexed single-photon generator. The method further includes coupling a first single photon emitted by the quantum emitter of the first single-photon source into the first optical cavity of the first single-photon source. The method further includes coupling a first single photon emitted by the quantum emitter of the first single-photon source into the first optical cavity. The method further includes coupling the first single photon from the first optical cavity into the bus waveguide of the multiplexed single-photon generator.

Description

BRIEF DESCRIPTION OF THE FIGURES

[0008] FIG. 1 is a perspective view of a single-photon source, in embodiments.

[0009] FIG. 2 is a side cutaway view of a center portion of the single-photon source of FIG. 1.

[0010] FIG. 3 is a top view of a single-photon source that incorporates the single-photon source of FIGS. 1 and 2 into a photonic integrated chip, in embodiments.

[0011] FIG. 4 is a top view of a time-multiplexed single-photon generator, in embodiments.

[0012] FIG. 5 shows a more detailed view of a single-photon source of the time-multiplexed single-photon generator of FIG. 4.

[0013] FIG. 6 illustrates a photon stream generated by the time-multiplexed single-photon generator of FIG. 4.

[0014] FIG. 7 is a top view of a frequency-multiplexed single-photon generator, in embodiments.

DETAILED DESCRIPTION

[0015] FIG. 1 is a perspective view of a single-photon source **100**, in accordance with some of the present embodiments. The single-photon source **100** includes an optical waveguide **104** that guides pump light **130** coupled into the waveguide **104** via an input face **126**. For clarity herein, the single-photon source **100** is oriented such that the waveguide **104** guides the pump light **130** in the +x direction of a right-handed Cartesian coordinate system **120**. The single-photon source **100** may also include a substrate **102** on which the waveguide **104** is positioned.

[0016] The single-photon source **100** includes a photonic crystal cavity **108** located adjacent to the optical waveguide **104**. The cavity **108** is created from an array of elliptical holes **114** formed in an optical thin film **112** that is deposited on top of (i.e., in the +z direction) the optical waveguide **104**. While the cavity **108** is shown in FIG. 1 forming an array of twelve holes **114** that are spaced along x, the cavity **108** may form a different number of holes **114**. While FIG. 1 shows the holes **114** as being elliptical, the holes **114** may alternatively be circular. The cavity **108** may alternatively be a different type of optical cavity or resonator known in the art.

[0017] FIG. 1 shows typical dimensions of the optical thin film **112**. In the example of FIG. 1, the thin film **112**, and therefore the photonic crystal cavity **108**, has the same width w (along y) as the optical waveguide **104**. Thus, the thin film **112** fully covers the waveguide **104** along y. The thin film **112** has a length l (along x) that is greater than the width w. The thin film **112** has a thickness t (along z) that is comparable to or less than the wavelength $\lambda_{\text{sub.p}}$ of the (monochromatic) pump light **130**. Typically, the thickness t is less than 1 μm . However, the thin film **112** and cavity **108** may have different dimensions without departing from the scope hereof.

[0018] The optical thinfilm **112** is formed from a solid-state material, such as a crystal. Examples of this solid-state material include, but are not limited to, titanium dioxide (TiO_2), calcium tungstate (CaWO_4), yttrium orthosilicate (Y_2SiO_5), yttrium orthovanadate (YVO_4), yttrium aluminum garnet ($\text{Y}_3\text{Al}_5\text{O}_{12}$), or any combination thereof. The substrate **102** is also a solid-state material, such as a polycrystalline or single-crystal semiconductor. Examples of the substrate material include, but are not limited to, silicon-on-insulator (SOI), lithium niobate, silicon nitride, aluminum nitride, strontium titanate, or any combination thereof.

[0019] The single-photon source **100** also includes at least one quantum emitter **110** that is embedded within the optical thin film **112**. The quantum emitter **110** may be an ion, molecule, color center (e.g., a nitrogen-vacancy center in diamond), crystal defect, quantum dot, or other type of quantum system or particle that can emit single photons in response to being optically pumped. In some embodiments, the thin film **112** is a crystal and the quantum emitter **110** is a rare-earth ion (e.g., erbium, ytterbium, neodymium, yttrium, praseodymium, europium, holmium, cerium, thulium, etc.) that substitutionally replaces one of the atoms of the crystal's lattice. The quantum emitter **110** may alternatively be interstitially embedded within the thin film **112** (e.g., trapped in a region between atoms of the crystal lattice).

[0020] The quantum emitter **110** is located between two of the holes **114** that are adjacent to each other. In FIG. 1, the quantum emitter **110** is centered (along y) with respect to the optical thin film **112**, and therefore the photonic crystal cavity **108**. The quantum emitter **110** is also centered (along z) with respect to the thickness t of the thin film **112**. However, the quantum emitter **110** need not be perfectly centered along x or y in this manner, as the quantum emitter **110** can still couple to the cavity **108** even when it is not perfectly centered. Also in FIG. 1, the quantum emitter **110** is centered (along x) between two of the holes **114** nearest the center of the cavity **108**. However, the quantum emitter **110** need not be centered along x in this manner. Furthermore, the quantum emitter **110** may be located between two other holes **114** that are adjacent to each other.

[0021] In some embodiments, the single-photon source **100** includes a plurality of quantum emitters **110** embedded within the optical thin film **112**. This plurality of quantum emitters **110** may all be the same type of quantum system (e.g., substitutional erbium ions). Alternatively, the plurality of quantum emitters **110** may include two or more different types of quantum systems (e.g., two different species of substitutional rare-earth ion). In some embodiments, the plurality of quantum emitters **110** has a density within the thin film **112** of 1 part-per-million (ppm) or less.

[0022] Pump light **130** propagating along the optical waveguide **104** evanescently couples into the optical thin film **112**, where it drives the quantum emitter **110**. Specifically, the frequency of the pump light **130** is chosen to excite an internal transition of the quantum emitter **110** (e.g., see FIG. 3). After excitation, the quantum emitter **110** spontaneously emits a single photon **132** that also propagates along the optical waveguide **104** in the +x direction. The single photon **132** exits the waveguide **104** via an output face **128**. Also propagating along the waveguide **104** is unabsorbed pump light **130'**, which also exits the waveguide **104** via the output face **128**.

[0023] In general, the quantum emitter **110** is located close enough to the photonic crystal cavity **108** that the photonic crystal cavity **108** modifies the final density of photonic states of the quantum emitter **110** when undergoing spontaneous decay. That is, the photonic crystal cavity **108** both modifies the spontaneous decay rate of the quantum emitter **110** and the spatial distribution of the spontaneous emission via the Purcell effect. Specifically, the photonic crystal cavity **108** has an optical resonance that coincides with the emission frequency of the spontaneous emission, which increases the spontaneous decay rate (i.e., decreases the spontaneous lifetime) and increases the probability that the quantum emitter **110** emits the single photon **132** such that it excites the optical resonance. The single photon **132** is then efficiently coupled from the photonic crystal cavity **108** into the optical waveguide **104**. By contrast, the quantum emitter **110**, in free space (e.g., in the absence of the photonic crystal cavity **108** or another structure that modifies that final density of

photonic states), spontaneously emits uniformly in all directions. In this latter case, most of the spontaneous emission cannot be collected and is therefore wasted. Thus, the photonic crystal cavity **108** improves the efficiency with which the single photon **132** is collected, guided along the waveguide **104**, and used for the application at hand.

[0024] The use of optical cavities to increase spontaneous decay rate and collection efficiency via the Purcell effect is well-known in the art and not limited to the photonic crystal cavity **108**. Rather, any type of optical cavity or photonic structure placed close enough to the quantum emitter **110** to modify its final density of photonic states may be used. To achieve significant Purcell enhancement, the single-photon source **100** operates in the “good emitter” regime (also known as the “bad cavity” regime) in which the spectral width of the spontaneous emission is narrower than the linewidth of the optical resonance. As an example, recent experiments using Er^{sup.3+} emitters in TiO₂ achieved spectral diffusion linewidths in the range of 0.2-0.6 GHz for optical resonances with linewidths in the range of 4-7 GHz [1]. A Purcell enhancement factor $F_{\text{sub.p}}$ up to several hundred, or more, can be obtained in practice.

[0025] There are several ways that the single-photon source **100** can be controlled to ensure that the optical resonance coincides with the wavelength of the single photon **132**. For example, the photonic crystal cavity **108** can be heated to change the size and spacings between the holes **114**. This heating may be performed by electrically driving a microheater located on the substrate **102** or otherwise in thermal contact with the cavity **108**. Alternatively, a strain can be applied to the cavity **108** (e.g., via a piezoelectric material) to change its refractive index and the length l . Alternatively, the transition frequency of the quantum emitter **110** can be modified via Stark tuning (i.e., applying an external electric field) or Zeeman tuning (i.e., applying an external magnetic field). As an example of Stark tuning, electrodes located near the quantum emitter **110** may be electrically driven to create the external electric field. These electrodes may be located in-plane with the thin film **112**, above the thin film **112**, or below the thin film **112**.

[0026] The photonic crystal cavity **108** is a one-dimensional photonic crystal in which each of the elliptical holes **114** has a major axis that is parallel to the y axis and a minor axis that is parallel to the x axis (alternatively, the major axis may be parallel to the x axis with the minor axis parallel to the y axis). The minor axes of the elliptical holes **114** are aligned, thereby forming an optical axis **118** of the cavity **108** that coincides with the propagation direction of light in the optical waveguide **104**. The cavity **108** has a first section in which the holes **114** are uniformly spaced (i.e., having a lattice constant that does not change along x) to form a first Bragg mirror **134** that reflects light at the resonant frequency of the cavity **108**. Note that this resonant frequency is similar to the frequency of the single photon **132**. Similarly, the cavity **108** has a second section in which the holes **114** are also uniformly spaced, thereby forming a second Bragg mirror **138** that reflects light at the resonant frequency. The cavity **108** has a middle section **136** in which the holes **114** are not uniformly spaced along x . The resulting irregularity of the lattice constant creates a defect that can be configured to transmit light at the resonant frequency therethrough. Alternatively, the holes **114** in the middle section **136** may be different sizes to vary the lattice constant along x . This structure of an optically transmissive section located between two mirror sections confines light along x . The elliptical shape of the holes **114** further confines light in the transverse y direction while the step changes in refractive index confines light along the transverse z direction. The result is that the cavity **108** confines light in all three spatial directions.

[0027] While FIG. 1 shows the photonic crystal cavity **108** being used in a two-sided transmission configuration in which the pump light **130** couples through the first Bragg mirror **134** and the single photon **132** couples through the second Bragg mirror **138**, the photonic crystal cavity **108** may alternatively be used in a one-sided reflection configuration. For example, the second Bragg mirror **138** can be made with more holes than the first Bragg mirror **134** to increase its reflectivity at the resonant frequency. In this case, the second Bragg mirror **138** reflects the single photon **132** into the $-x$ direction, where it will couple through the first Bragg mirror **134** and exit the optical

waveguide **104** via the input face **126**.

[0028] The quantum emitter **110** is located near the center of the photonic crystal cavity **108** such that it spatially overlaps the largest electric field of the cavity's resonant mode. This spatial overlap ensures strong coupling between the quantum emitter **110** and the cavity **108**. The middle section **136** can be configured such that longitudinally (i.e., along x), the largest electric field is at the center-most antinode of the resonant mode (i.e., the antinode that occurs between the two center-most holes **114** of the middle section **136**). The electric field of the resonant mode is linearly polarized along y. Accordingly, when the quantum emitter **110** has a dipole moment, the coupling strength between the quantum emitter **110** and the cavity **108** will depend on the angle between the dipole moment and the y axis, with the strongest coupling occurring when the dipole moment is aligned along y.

[0029] The photonic crystal cavity **108** may be designed, fabricated, and used using techniques known in the art. For example, additional details about how to design the photonic crystal cavity **108** for Er emitters can be found in Refs. [2] and [3].

[0030] FIG. 2 is a side cutaway view of a center portion of the single-photon source **100** of FIG. 1, taken along the optical axis **118** of the photonic crystal cavity **108**. FIG. 2 shows one example of a heterostructure that may be grown and used as the optical thin film **112** of FIG. 1. The quantum emitter **110** is located within a doped layer **212** that is sandwiched between an underlying undoped layer **204** and an overlaying undoped layer **202**. The undoped layers **202** and **204** act as buffer layers that protect the quantum emitter **110**. The layers **202**, **204**, and **212** may have the same thickness (as measured along z) or different thicknesses. For example, each of the layers **202**, **204**, and **212** may have a thickness of 7.5 nm, for a total thickness of approximately 22 nm. However, each of the layers **202**, **204**, and **212** may have a different thickness.

[0031] In some embodiments, the layers **202**, **204**, and **212** are TiO₂ and the quantum emitter **110** is a trivalent Er³⁺ rare-earth ion. In these embodiments, the optical waveguide **104** may be fabricated from silicon. This silicon may be the handle layer of an SOI wafer. These advantages are particularly advantageous when the single photon **132** has a wavelength within, or near, the telecom C-band between 1530 and 1565 nm. In this band, silicon is highly transmissive. However, for other wavelengths, other types of materials may be used for the waveguide **104** (e.g., silicon nitride (SiN), lithium niobate (LiNbO₃), aluminum nitride (AlN), etc.). Similarly, the waveguide **104** and layers **202**, **204**, and **212** may be fabricated on a wafer of a different material (e.g., SiN, lithium nickel dioxide (LNO), etc.). Additional details about fabricating the layers **202**, **204**, and **212**, especially for the case of Er ions embedded within TiO₂ thin films, can be found in Refs. [1-5] as well as International Publication WO 2023091546A1 and U.S. Patent Application Pub. No. 2022/0136133. Each of these publications is incorporated herein by reference in its entirety.

[0032] FIG. 3 is a top view of a single-photon source **300** that incorporates the single-photon source **100** of FIGS. 1 and 2 into a photonic integrated chip (PIC). The single-photon source **300** includes an input coupler **302** and an input waveguide **308** that guides the pump light **130** from the input coupler **302** to the single-photon source **100**. The single-photon source **300** also includes an output coupler **304** and an output waveguide **310** that guides the single photon **132** from the single-photon source **100** to the output coupler **304**. The input coupler **302**, input waveguide **308**, output waveguide **310**, and output coupler **304** are fabricated on top of, or inside of, a top surface of the substrate **102**, similar to the single-photon source **100**.

[0033] The input coupler **302** connects to an input optical fiber **318** to form an input interconnect **312**. Similarly, the output coupler **304** connects to an output optical fiber **328** to form an output interconnect **322**. With the interconnects **312** and **322**, the single-photon source **300** may be used with other optical components located “off-chip.” In the example of FIG. 3, the input coupler **302** is a surface grating coupler that uses diffraction to couple with the input optical fiber **318**. The output coupler **304** is also shown as a surface grating coupler. Alternatively, each of the couplers

302 and **304** may be configured for end-fire coupling or adiabatic coupling. The optical fibers **318** and **328** may be conventional single-mode fiber (e.g., SMF-28), polarization-maintaining fiber, photonic crystal fiber, or another type of optical fiber known in the art.

[0034] In some embodiments, the single-photon source **300** is fabricated from cryogenic-compatible materials, which advantageously allows the single-photon source **300** to be placed in a cryogenic environment (e.g., temperatures as low as 10 mK, or less) while other optical components (e.g., a pump laser that generates the pump light **130**) are located at room temperature outside of the cryogenic environment. In this case, the optical fibers **318** and **328** may be used to guide the pump light **130** and single photon **132**, respectively, between these two environments. To ensure that the interconnects **312** and **322** are robust enough to survive cryogenic temperature cycling, the optical fibers **318** and **328** may be connected to the couplers **302** and **304** with photonic wire bonding, butt-coupling, or facet-attached microlenses.

[0035] In FIG. 3, the waveguides **308** and **310** are curved such that the couplers **302** and **304** are located along the same edge of the substrate **102**. However, the waveguides **308** and **310** may be shaped differently. Accordingly, the couplers **302** and **304** may be located on different edges of the substrate **102**. Although not shown in FIG. 3, the single-photon source **300** may include additional optical components incorporated into the substrate **102** without departing from the scope hereof. Examples of such optical components include, but are not limited to, resonators, filters, splitters/combiners, additional waveguides, and nonlinear optical elements. The single-photon source **300** may also include electronic elements incorporated within or on the substrate **102**. Examples of such electronic components include, but are not limited to, microheaters, electrodes, wires (i.e., electrically conductive paths), and photodetectors.

[0036] FIG. 4 is a top view of a time-multiplexed single-photon generator **400** that generates a photon stream **428** of single photons **420**, in accordance with some of the present embodiments. The single-photon generator **400** includes N of the single-photon source **100** of FIG. 1, shown in FIG. 4 as a first single-photon sources **412(1)** through an N^{sup.th} single-photon source **412(N)**. FIG. 5 shows a more detailed view of the first single-photon source **412(1)** of FIG. 4. FIG. 6 illustrates the photon stream **428** of FIG. 4 in more detail. FIGS. 4-6 are best viewed together with the following description.

[0037] The time-multiplexed single-photon generator **400** includes a 1:N optical switching network **408** and an input waveguide **406** that couples pump light **404** to an input port of the optical switching network **408** (labeled “IN” in FIG. 4). The optical switching network **408** has N output ports (labeled “OUT(1)” through “OUT(N)” in FIG. 4) that are respectively connected to N intermediate waveguides **410(1)** through **410(N)**. A controller **426** electrically controls the optical switching network **408**, via an electrical bus **424**, to connect the input port to any one of the N output ports. For example, when the controller **426** controls the optical switching network **408** to connect the input port to the i^{sup.th} output port, then the pump light **404** exits the optical switching network **408** via the i^{sup.th} output port and propagates along the i^{sup.th} intermediate waveguide **410(i)**. Here, i is an integer with a value between 1 and N, inclusive.

[0038] The N single-photon sources **412** are co-linear, i.e., they are aligned such that the optical axis **118** of each of the N photonic crystal cavities **108** coincides with the same line (which is parallel to x in FIGS. 4 and 5). To evanescently couple the pump light **404** into the single-photon sources **412**, each intermediate waveguide **410(i)** forms a curve, at the end of which is a linear segment **422(i)** that runs parallel to the single-photon source **412(i)**. For example, FIG. 5 shows the first intermediate waveguide **410(1)** forming a first linear segment **422(1)** that runs parallel to a first photonic crystal cavity **108(1)** of the first single-photon source **412(1)**. The linear segment **422(1)** is displaced from the single-photon source **412(1)** in the +y direction to form a gap **434** therebetween that is small enough that the pump light **404** can evanescently couple into the single-photon source **412(1)**. Specifically, the size of the gap **434** (along y) is comparable to or less than the wavelength $\lambda_{sub.p}$ of the pump light **404**. Typically, the size of the gap **434** is in the range of

100-400 nm, with the size determining whether the photonic crystal cavity **108(1)** is overcoupled, undercoupled, or critically coupled with the photonic crystal cavity **108(1)**. Furthermore, the length of the linear segment **422(1)**, along x, that is adjacent to the photonic crystal cavity **108(1)** is much longer than $\lambda_{\text{sub.p}}$, again to ensure that the pump light **404** can evanescently couple into the photonic crystal cavity **108(1)**. The geometry shown in FIG. 5 applies to all N of the linear segments **422** and single-photon sources **412** of FIG. 4.

[0039] The time-multiplexed single-photon generator **400** also includes a bus waveguide **414** that runs adjacent to the single-photon sources **412** and is displaced from the single-photon sources **412** in the $-y$ direction. As shown in FIG. 5, a gap **436** between the bus waveguide **414** and the single-photon sources **412** is small enough that single photons **420** generated by any of the N single-photon sources **412** can evanescently couple into the bus waveguide **414**. Specifically, the size of the gap **436** (along y) is comparable to or less than the wavelength $\lambda_{\text{sub.S}}$ of the single photons **420**. Typically, the size of the gap **436** is in the range of 100-400 nm, with the size determining whether the photonic crystal cavity **108(1)** of the single-photon source **412(1)** is overcoupled, undercoupled, or critically coupled with the bus waveguide **414**.

[0040] FIG. 5 shows how a first quantum emitter **110(1)** of the first single-photon source **412(1)** can emit either a single photon **420''** that propagates along the bus waveguide **414** in the $+x$ direction or a single photon **420'** that propagates along the bus waveguide **414** in the $-x$ direction. As shown in FIG. 4, the time-multiplexed single-photon generator **400** may also include a mirror **416** that is located at one end of the bus waveguide **414** and retroreflects the single photon **420'** such that it then propagates along the bus waveguide **414** in the $+x$ direction. Thus, the mirror **416** ensures that all single photons emitted by all of the N single-photon sources **412** propagate along the bus waveguide **414** in the same direction. The mirror **416** may be a Bragg reflector, Sagnac loop, or another type of mirror or retroreflector known in the art.

[0041] The time-multiplexed single-photon generator **400** advantageously generates single photons **420** at a rate that is higher than what can be achieved using only one of the single-photon sources **412** (i.e., without time-multiplexing). For example, assume that all of the single photons **420** have the same temporal width $T_{\text{sub.S}} = 1 \mu\text{s}$ and that the emission rate of single photons **420** from the first photonic crystal cavity **108(1)** is $T_{\text{sub.C}}^{-1} = (10 \mu\text{s})^{-1} = 10 \text{ kHz}$, where $T_{\text{sub.C}}$ is the average emission time of single photons **420** from the photonic crystal cavity **108(1)**. In this case, there is, on average, about $(T_{\text{sub.C}} - T_{\text{sub.S}}) \approx 9 \mu\text{s}$ of “dead” time between two sequential single photons **420** emitted by the single-photon source **412(1)**. During this dead time, up to $(T_{\text{sub.C}} - T_{\text{sub.S}})/T_{\text{sub.S}} = 9$ additional single photons **420** can be generated and emitted without temporal overlap (as needed to ensure their indistinguishability). These additional single photons **420** can be generated with additional single-photon sources **412**. To prevent temporal overlap, the maximum number N of single-photon sources **412** that can be used is approximately $N \approx T_{\text{sub.C}}/T_{\text{sub.S}}$.

[0042] FIG. 6 illustrates one example of the photon stream **428** of FIG. 4. In this example, the single photons **420** all have the same temporal width $T_{\text{sub.S}}$ and wavelength. The first single-photon source **412(1)** of FIG. 4 emits a first single photon **420(1)**, after which it is pumped again. During this pumping and subsequent decay, the second single-photon source **412(2)** of FIG. 4 emits a second single photon **420(2)** at a time $T_{\text{sub.P}}$ after the first single photon **420(1)**. After this emission, the second single-photon source **412(2)** is pumped again. This process of spontaneous emission and pumping continues through all N single-photon sources **412**, thereby extending the photon stream **428** to an $N_{\text{sup.th}}$ single photon **420(N)** emitted by the $N_{\text{sup.th}}$ single-photon source **412(N)**. After time $T_{\text{sub.C}}$ has passed since the first single photon **420(1)** was emitted, the first single-photon source **412(1)** then emits another single photon **420(N+1)**. Single-photon generation then continues, as described, cyclically through the N single-photon sources **412(1) . . . 412(N)** to generate single photons **420(N+2)** through **420(2N)**, single photons **420(2N+1)** through **420(3N)**, and so on.

[0043] Since spontaneous emission is a random process, the time at which a quantum emitter **110** emits a single photon is random. Similarly, the time at which a photonic crystal cavity **108** emits is also random. Accordingly, $T_{sub.P}$ and $T_{sub.C}$ are shown in FIG. **6** as average times. In practice, due to the random nature of the spontaneous emission, the spacing between single photons will vary and will likely not be periodic. For this reason, the sequence of single photons **420** is referred to as a “stream” instead of a “pulse train.”

[0044] To implement cyclical single-photon generation with the N single-photon sources **412**, the controller **426** controls the optical switching network **408** to route the pump light **404** to each of the N single-photon sources **412** at the appropriate time. In one embodiment, the controller **426** controls the optical switching network **408** to output pump light **404** sequentially through the N output ports over the time $T_{sub.C}$, thereby sequentially pumping the N single-photon sources **412** one-at-a-time. This control may be cyclical, i.e., pump light **404** is outputted from the first output port following pump light **404** outputted from the $N_{sup.th}$ output port.

[0045] The optical switching network **408** may be fabricated using any kind of active (i.e., electrically controlled) optical switch known in the art. Examples of such optical switches include, but are not limited to, Mach-Zehnder interferometers and ring switches. In another embodiment, a passive delay network is used for the optical switching network **408**. In this embodiment, the pump light **404** is delayed (e.g., via an appropriate length of an optical waveguide or optical fiber) based on which of the N single-photon sources **412** it pumps.

[0046] The time-multiplexed single-photon generator **400** may include one or more additional components integrated with the substrate **402**. For example, FIG. **4** shows the generator **400** with an optical filter **418** that spectrally filters the photon stream **428**. The filter **418** may, for example, block residual pump light **404** that couples into the bus waveguide **414**. The filter **418** may also be used to spectrally select single photons **420** lying within a certain spectral band. The filter **418** may be a microresonator, Mach-Zehnder interferometer, or other type of optical component that performs spectral filtering. In the example of FIG. **4**, the photon stream **428** exits the filter **418** along an exit waveguide **432**. Light rejected by the filter **418** may be coupled into an on-chip beam dump or another waveguide that guides the rejected light to an optical interconnect for dumping this rejected light off-chip. In another embodiment, the generator **400** excludes the filter **418**, in which case the bus waveguide **414** and exit waveguide **432** are the same. In this embodiment, filtering may be performed off-chip (e.g., via a fiber-optic-based filter) or not at all.

[0047] Although not shown in FIG. **4**, the time-multiplexed single-photon generator **400** may include the input coupler **302** of FIG. **3** to form the input interconnect **312**. Additionally or alternatively, the generator **400** may include the output coupler **304** of FIG. **3** to form the output interconnect **322**. Thus, the generator **400** may be used with optical fibers, such as to enable cryogenic operation and operation with additional external components (e.g., a pump laser).

[0048] FIG. **7** is a top view of a frequency-multiplexed single-photon generator **700** that generates a photon stream **728** of single photons **720**, in accordance with some of the present embodiments. The frequency-multiplexed single-photon generator **700** is similar to the time-multiplexed single-photon generator **400** of FIG. **4** except that it includes N single-photon sources **712** that are configured to emit single photons **720** at different frequencies. By contrast, the single-photon sources **412** of FIG. **4** are all configured to emit single photons **420** at the same frequency. The single-photon sources **712** may be configured to emit single photons at different frequencies using techniques described above (e.g., Stark-shift tuning via electrodes). Each of the single-photon sources **712** includes a photonic crystal cavity similar to the photonic crystal cavity **108** described above. As an alternative to Stark or Zeeman tuning (or in addition to these tuning mechanisms), the photonic crystal cavities of the single-photon sources **712** may be fabricated with different lattice constants so that they resonate at different frequencies.

[0049] Since single photons **720** emitted from different single-photon sources **712** are distinguishable based on frequency, it is not necessary that such single photons **720** be temporally

distinguishable. Accordingly, the frequency-multiplexed single-photon generator **700** may be operated such that at least some of the single photons **720** are temporally overlapped, as shown in FIG. 7, and therefore the rate of the photon stream **728** can exceed the rate of the photon stream **428** of FIG. 4, even though only one of the single-photon sources **712** can be pumped at any given time. Such temporal overlap can be achieved because the time needed to pump any one of the single-photon sources **712** is short compared to its spontaneous lifetime. Once coupled off-chip, the single photons **720** can be demultiplexed or frequency converted to the same frequency.

[0050] While the above embodiments are shown implemented using PIC technology, it should be appreciated that each embodiment herein need not be implemented entirely as a PIC-based device fabricated on the substrate **402**. For example, the optical switching network **408** can be constructed using fiber-optic switches and components. In this case, each of the output ports of the optical switching network **408** can be coupled to its respective intermediate waveguide **410(i)** via an optical interconnect similar to the optical interconnects **312** and **322** shown in FIG. 3.

[0051] Changes may be made in the above methods and systems without departing from the scope hereof. It should thus be noted that the matter contained in the above description or shown in the accompanying drawings should be interpreted as illustrative and not in a limiting sense. The following claims are intended to cover all generic and specific features described herein, as well as all statements of the scope of the present method and system, which, as a matter of language, might be said to fall therebetween.

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Claims

1. A multiplexed single-photon generator, comprising: a plurality of single-photon sources, each of the plurality of single-photon sources comprising an optical cavity and a quantum emitter coupled to the optical cavity; an optical switching network controllable to couple pump light into the optical cavity of any one of the plurality of single-photon sources; and a bus waveguide coupled to the optical cavity of each of the plurality of single-photon sources.
2. The multiplexed single-photon generator of claim 1, further comprising a retroreflector located at an end of the bus waveguide.
3. The multiplexed single-photon generator of claim 1, further comprising an optical filter coupled to the bus waveguide and configured to couple unabsorbed pump light out of the bus waveguide.
4. The multiplexed single-photon generator of claim 1, the optical cavity comprising a photonic crystal cavity.
5. The multiplexed single-photon generator of claim 1, the optical cavity being evanescently coupled to the bus waveguide.
6. The multiplexed single-photon generator of claim 1, wherein: the optical switching network comprises a plurality of output ports; and each of the plurality of output ports is evanescently coupled to the optical cavity of a respective one of the plurality of single-photon sources.
7. The multiplexed single-photon generator of claim 1, wherein: the quantum emitter of each of the

plurality of single-photon sources, in response to being pumped by the pump light, spontaneously decays to emit a single photon having a respective one of a plurality of center frequencies; and all of the plurality of center frequencies are the same.

8. The multiplexed single-photon generator of claim 1, wherein: the quantum emitter of each of the plurality of single-photon sources, in response to being pumped by the pump light, spontaneously decays to emit a single photon having a respective one of a plurality of center frequencies; and at least two of the plurality of center frequencies differ from each other.

9. The multiplexed single-photon generator of claim 1, further comprising a substrate; wherein the plurality of single-photon sources and the bus waveguide are located on or within the substrate.

10. The multiplexed single-photon generator of claim 9, the substrate comprising silicon-on-insulator, lithium niobate, silicon nitride, aluminum nitride, strontium titanate, or any combination thereof.

11. The multiplexed single-photon generator of claim 1, the quantum emitter comprising a thin film of a solid-state material doped with one or more rare-earth ions.

12. The multiplexed single-photon generator of claim 11, the one or more rare-earth ions comprising a plurality of rare-earth ions having a density of 1 ppm or less.

13. The multiplexed single-photon generator of claim 11, each of the one or more rare-earth ions being an erbium ion, a praseodymium ion, a neodymium ion, a ytterbium ion, or a europium ion.

14. The multiplexed single-photon generator of claim 11, the solid-state material comprising titanium dioxide (TiO₂), calcium tungstate (CaWO₄), yttrium orthosilicate (Y₂SiO₅), yttrium orthovanadate (YVO₄), yttrium aluminum garnet (Y₃Al₅O₁₂), or any combination thereof.

15. A method for single-photon generation, comprising: coupling a first pump pulse into the optical switching network of the multiplexed single-photon generator of claim 1; controlling the optical switching network to couple the first pump pulse into the optical cavity of a first single-photon source of the plurality of single-photon sources of the multiplexed single-photon generator; coupling a first single photon emitted by the quantum emitter of the first single-photon source into the first optical cavity; and coupling the first single photon from the first optical cavity into the bus waveguide of the multiplexed single-photon generator.

16. The method of claim 15, further comprising: coupling a second pump pulse into the optical switching network; controlling the optical switching network to couple the second pump pulse into a second single-photon source of the plurality of single-photon sources, the second single-photon source being different from the first single-photon source; coupling a second single photon emitted by the quantum emitter of the second single-photon source into a second optical cavity of the second single-photon source; and coupling the second single photon from the second optical cavity into the bus waveguide.

17. The method of claim 16, wherein said coupling the second pump pulse occurs such that a temporal duration between the first single photon and the second single photon is greater than one or both of a first temporal width of the first single photon and a second temporal width of the second temporal width.

18. The method of claim 16, wherein said coupling the second pump pulse occurs such that a temporal duration between the first single photon and the second single photon is less than one or both of a first temporal width of the first single photon and a second temporal width of the second temporal width.

19. The method of claim 16, wherein: the first single photon has a first center frequency; and the second single photon has a second center frequency different from the first center frequency.

20. The method of claim 19, further comprising frequency demultiplexing the first single photon and the second single photon.
